



Home Loan Advisor

A Consumer-Facing Mortgage Risk Intelligence MVP

DATA 622 | Machine Learning in Production | Spring 2026

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The Problem

\$300K+

Average U.S. mortgage — the largest financial decision most people will ever make

0 tools

Consumer-facing tools to assess personal mortgage default risk before committing

The Gap

Banks have sophisticated ML models to evaluate borrowers — but consumers have nothing equivalent. Home Loan Advisor flips the perspective: it empowers the borrower to make data-driven decisions before committing to a mortgage, using a model trained on real Freddie Mac data.

Who Is This For?

First-Time Homebuyers

Unfamiliar with how mortgage risk is assessed — want guidance before approaching a lender.



Refinancing Homeowners

Want to understand how their current profile compares to historical default patterns.



Financial Advisors

Need a quick, visual tool to walk clients through mortgage readiness scenarios.

Data Source



Freddie Mac Single Family Loan-Level Dataset

Loan-level origination & performance data on millions of U.S. mortgages (1999–present).

MVP scope: 2018–2020 vintages | 150,000 loan sample | 4.31% positive class rate

Feature	Description	Type
borrower_credit_score	FICO credit score at origination	Numeric
original_dti	Debt-to-income ratio at origination	Numeric
original_ltv	Loan-to-value ratio at origination	Numeric
original_interest_rate	Interest rate at origination	Numeric
property_state, property_type	State, single-family/condo/etc.	Categorical
ever_delinquent	Target: loan became 90+ days delinquent	Binary

ML Approach

Binary Classification — does a borrower ever become 90+ days delinquent?

Logistic Regression

Interpretable baseline.
Coefficients show which features
increase/decrease default risk.
Trained on 150K loans.

MVP Model

Random Forest

Captures non-linear interactions.
Built-in variable importance
for the feature influence panel.

Post-MVP

XGBoost

Best performer on tabular
financial data. Hyperparameter-
tuned via cross-validation.

Post-MVP

All models implemented with tidymodels in R | Serialized as .rds for Shiny inference

App Features — 5 Tabs



Overview

Executive summary with key stats (150K loans, 4.31% positive rate, AUC 0.755), scope, and limitations



My Profile

Input form with risk estimate, risk tier, recommendation panel, top drivers chart, and sample comparison table



Scenario Simulator

What-if sliders for income, loan amount, property value, DTI, and rate changes — with baseline vs scenario comparison



Model Evidence

ROC curve, confusion matrix, calibration plot, placeholder audit, borrower demo cases reproduced in R



Project Review

Proposal alignment, what the app fixes vs. the notebook demo, important limitations, and transparency notes

Model Performance

150K

Loans Trained

Freddie Mac 2018–2020
stratified sample

0.755

Held-Out AUC

Correctly ranks risky vs.
safe borrowers 76% of the time

0.753

Mean CV AUC

Stable across cross-validation
folds — not overfitting

4.31%

Positive Class

Only ~4% of loans default —
heavy class imbalance

In Plain English

If you lined up 100 past borrowers and asked this model to sort them from most to least likely to default, it would get the ranking right about 76 out of 100 times. We emphasize AUC over accuracy because with only 4.3% defaults, a naive model scores 96% but catches zero defaults.

Model Evidence Included

- ✓ ROC curve with AUC = 0.755
- ✓ Confusion matrix at threshold 0.10
- ✓ Calibration plot (predicted vs observed)
- ✓ Freddie Mac placeholder audit table
- ✓ Borrower demo cases reproduced in R

Addressing Feedback



Present accuracy for non-ML audience

Recommendation panel uses plain language ("relatively favorable"). AUC emphasized over accuracy because 96% accuracy is misleading with 4.3% default rate.



Cite references in text

Data source, methodology, and limitations are documented inline throughout the Overview and Project Review tabs with explicit scope notes.



Imputation for missing user data

Income used for affordability context only (not in model). App notes which fields come from Freddie Mac vs. user input. Placeholder values (9999) audited.



Sensitivity plot / scenario analysis

Scenario Simulator tab with 5 adjustment sliders, baseline vs. scenario comparison, predicted risk shift chart, and values-before-and-after table.

MVP Status & Next Steps

MVP Delivered

- ✓ Trained on real Freddie Mac data
- ✓ 150K loans, 2018–2020 vintages
- ✓ Logistic regression (AUC 0.755)
- ✓ Input form with all borrower fields
- ✓ Real-time risk estimate + tier
- ✓ Scenario simulator with 5 sliders
- ✓ Feature importance (log-odds)
- ✓ ROC, confusion matrix, calibration
- ✓ Project review + limitations tab
- ✓ Deployed on shinyapps.io

Post-MVP Roadmap

- Expand to 2018–2022 vintages
- Add Random Forest & XGBoost
- Sensitivity curve plots
- Usability testing (3–5 peers)

Honest Limitations

MVP baseline, not production underwriting
Income used for affordability only (not in model)
Risk tiers are prototype, not business-calibrated
Scope: 2018–2020, not full 2018–2022 vision

References

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